

Proposed Solution

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Solution Overview

Smart Lender is a machine-learning-based web application that predicts whether a loan application is likely to be approved or rejected, based on applicant attributes such as income, credit history, employment status, and loan amount requested.

How It Works

- 1. Historical loan application data is collected and cleaned.
- 2. Exploratory Data Analysis (EDA) identifies key patterns and relationships affecting approval.
- 3. Data is preprocessed: missing values handled, categorical variables encoded, dataset balanced, features scaled.
- 4. Multiple ML classification models are trained and compared.
- 5. The best-performing model is saved and integrated into a Flask web application.
- 6. Users submit their details via a form and receive an instant prediction.

Unique Value Proposition

- Instant, self-service eligibility check — no waiting for manual review.
- Consistent, data-driven decisioning logic.
- Simple, lightweight architecture that's easy to extend or retrain on new data.

Target Users

- Individuals applying for personal/home loans who want a quick pre-check.
- Loan officers seeking a first-pass automated filter.